
RESEARCH PROFILE

Computer Science graduate from the University of Warwick with research and professional experience in machine-learning-based predictive modelling, cyber-risk analysis, distributed trading systems, and real-time monitoring. My work has involved modelling noisy industrial data, analysing cybersecurity risks in vehicle-to-grid systems, and developing latency-sensitive financial infrastructure with asynchronous coordination and failure monitoring. These experiences have motivated my interest in risk intelligence for complex engineering, energy, and decentralised financial systems. I am particularly interested in combining stochastic modelling and uncertainty quantification with machine learning and distributed-systems security to detect anomalous or adversarial behaviour, quantify rare but consequential risks, and transform heterogeneous operational and transaction data into robust decision support.

RESEARCH INTERESTS

Risk Intelligence; Stochastic Modelling and Uncertainty Quantification; AI for Security; Cyber-Physical and Decentralised Financial Systems; Anomaly, Adversarial-Behaviour, and Rare-Event Detection; Risk-Aware Decision-Making.

EDUCATION

- **University of Warwick** Coventry, UK
MSc in Predictive Modelling and Scientific Computing, Merit Oct. 2023 – Nov. 2024
Dissertation: Leveraging Data-Driven Insights for Enhanced Aluminium Pressing Quality
Supervisor: Dr Ishwar Kapoor
- **University of Warwick** Coventry, UK
BSc in Computer Science, 2:1 Honours Sep. 2020 – Jul. 2023
Dissertation: Location Extraction and Movement Prediction for Trajectory
Supervisor: Prof. Nathan Griffiths

RESEARCH EXPERIENCE

- **Cyber Risks and Threats Associated with Vehicle-to-Grid Systems** Coventry, UK
WMG, University of Warwick Jun. - Oct. 2022
Student Researcher
 - Conducted a structured review of cybersecurity risks arising from Vehicle-to-Grid integration, focusing on threats to communication, authentication, grid interaction, and ancillary-service operation.
 - Analysed how cyber threats, operational dependencies, and connected infrastructure could affect the reliability and resilience of V2G-enabled energy systems.
 - Organised identified vulnerabilities, attack surfaces, and potential consequences into a structured risk-identification framework.
- **Leveraging Data-Driven Insights for Enhanced Aluminium Pressing Quality** Coventry, UK
WMG, University of Warwick Sep. 2023 - Oct. 2024
MSc Dissertation Researcher
 - Investigated whether aluminium-alloy hardness could be predicted from chemical composition and industrial processing profiles using supervised machine-learning models.
 - Developed and evaluated DNN, SVR, decision-tree, and XGBoost regression models using industrial production and quality-control data.
 - Applied Local Outlier Factor to investigate anomalous observations in the ageing process and evaluated their impact on predictive performance.
- **Location Extraction and Movement Prediction from GPS Trajectory** Coventry, UK
Department of Computer Science, University of Warwick Sep. 2022 - Jun. 2023
Undergraduate Dissertation Researcher
 - Developed clustering-based location-extraction models using k-means and DBSCAN, and evaluated them on the GeoLife trajectory dataset.
 - Built an LSTM-based movement-prediction pipeline for heterogeneous spatio-temporal trajectory data.

- Reduced clustering runtime by approximately 10x through trajectory segmentation, preprocessing, and implementation optimisation.

- **Construction Data Analysing and Modelling**

Coventry, UK
Oct. 2022 - Mar. 2023

Warwick AI Society, University of Warwick
Student Researcher

- Developed a Python NLP pipeline combining sentiment analysis, topic modelling, and feature ranking to analyse construction and urban-development text data.
- Implemented API-based data collection and contributed to a Chrome-extension interface for presenting extracted topics and sentiment indicators.

SELECTED INDUSTRY EXPERIENCE

- **DP Consulting**

Warwickshire, UK

Software Engineer - Trading Systems

Jun. 2025 - Present

Architected and deployed a production-grade distributed MT5 copy-trading platform.

- Designed a distributed trading platform supporting more than 50 concurrent accounts across multiple hosts, with real-time signal replication below 100 ms.
- Developed state-machine orchestration and CAS-based synchronisation to mitigate inconsistent execution states, increasing coordinated execution success from 60% to 95%.
- Built real-time monitoring and event-reporting infrastructure for identifying execution failures, latency anomalies, and account-state inconsistencies.

- **Outlier**

Remote

AI Data Specialist

Oct. 2024 - Present

LLM evaluation and training-data projects in coding, mathematics, and multilingual reasoning.

- Contribute to LLM evaluation projects assessing coding and mathematical-reasoning capabilities.
- Participate in supervised fine-tuning and reinforcement-learning-from-human-feedback data projects.
- Serve as a senior reviewer across coding and mathematics projects and deliver training webinars for contributors.

- **IShangu Cyber Security Technology Co., Ltd**

Zhengzhou, China

Information Systems Intern

Jul. - Aug. 2023

Managed and optimised databases on Kylin OS, performed advanced Linux server administration and implemented virtualisation technologies.

- Managed and optimized MySQL, Oracle, and Kingbase8 databases within the Kylin OS, improving performance and ensuring data integrity.
- Implemented virtualization technologies and Linux system administration, including LVM disk expansion, new disk mounting, NFS configuration, and common Linux commands for system optimization and efficient infrastructure management.

- **Saint Laurent Consulting**

Remote

Business Intelligence Engineer Intern

Oct. 2022 - Jan. 2023

Client analytics project for Amazon fulfilment centres

- Engineered an automated ETL pipeline using Python, BeautifulSoup, and Pandas for web scraping, with data efficiently stored in a PostgreSQL database, ensuring accurate and structured data management.
- Built and deployed an automatically updated analytics dashboard using Tableau and Flask, integrating REST API data and AWS-based scheduled processing.

TECHNICAL SKILLS

- **Programming:** Python, Java, SQL, C++, Go
- **Machine Learning and Data Analysis:** PyTorch, scikit-learn, Keras, regression, clustering, anomaly detection, NLP, time-series and trajectory modelling
- **Security and Risk Analysis:** cyber-risk identification, threat analysis, anomaly detection, failure monitoring
- **Distributed Systems:** asynchronous programming, message queues, state-machine orchestration, distributed coordination, REST APIs, FastAPI, NATS JetStream
- **Data and Infrastructure:** PostgreSQL, TimescaleDB, MySQL, Docker, Linux, Git, AWS